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Paleomagnetism of Middle Proterozoic (1.01 to 1.08 Ga) mafic dykes in southeastern Bahia State—São Francisco Craton, Brazil

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ABSTRACT

Paleomagnetic results from profuse Middle–Late Proterozoic mafic dyke swarms in southeastern Bahia State (São Francisco Craton) yield either an easterly direction with high upward inclination or a westerly direction with high downward inclination isolated during AF and/or thermal treatments. Thermal demagnetization behavior and thermochronologic and petrologic considerations indicate that these remanent components originated as primary TRM's. Four groups of directions were distinguished from dykes in spatially distinct areas: Ilheus normal polarity ($D = 60.0°$; $I = -68.8°$; $\alpha_{95} = 2.6°$; $N = 17$), Olivença normal polarity ($D = 82.4°$; $I = -71.0°$; $\alpha_{95} = 5.1$; $N = 31$), Itajú do Colônia ($D = 99.0$; $I = -71.9$; $\alpha_{95} = 5.9°$; $N = 23$) and Olivença reversed polarity ($D = 298.8°$; $I = 60.7°$; $\alpha_{95} = 6.4°$; $N = 18$), which yield paleomagnetic poles located at 100.4°E; 30.3°N (I_N), 107.0°E; 16.1°N (O_N), 111.0°E; 7.7°N (IC) and 280.2°E; 17.0°N (O_R), respectively. These poles define an APW path for the São Francisco Craton between the time interval 1.01–1.08 Ga which is characterized by at least two polarity intervals.

1. Introduction

Published paleomagnetic and $^{40}Ar/^{39}Ar$ studies of Precambrian rocks from the Amazonian craton have helped to establish the thermal and tectonic evolution of this South American cratonic area [1–4]. To date however, no paleomagnetic study concerning well-dated Precambrian rocks from the São Francisco craton has been published.

This is the first of two papers presenting new paleomagnetic and $^{40}Ar/^{39}Ar$ data from numerous Middle Proterozoic mafic dyke swarms in southeastern Bahia State (São Francisco Craton). The detailed paleomagnetic results from these dykes will be discussed herein while the second paper [5] presents the $^{40}Ar/^{39}Ar$ dating of these dykes and of nearly contemporaneous magnetizations from the Kalahari Craton, and discusses tectonic implications for the Brasiliano and Pan-African mobile belts of South America and southern Africa.

2. Geological setting

The dyke swarms studied are located in the eastern part of Bahia State, Brazil. The area belongs to the São Francisco Craton which is an extension of the Congo Craton in west Africa (Fig. 1). Restored to a pre-Atlantic configuration these cratons are jointly encircled by Late Proterozoic mobile belts, demonstrating their coherent stable tectonic character during the Brasiliano/Pan African Orogeny. Within these cratons the crustal evolution is characterized by an Archean to Early Proterozoic polymetamorphic basement overlain by subparallel, elongated Middle Proterozoic belts associated with platform covers (Espinhaço of the São Francisco Craton and Mayumbe of the Congo Craton) [8–10].

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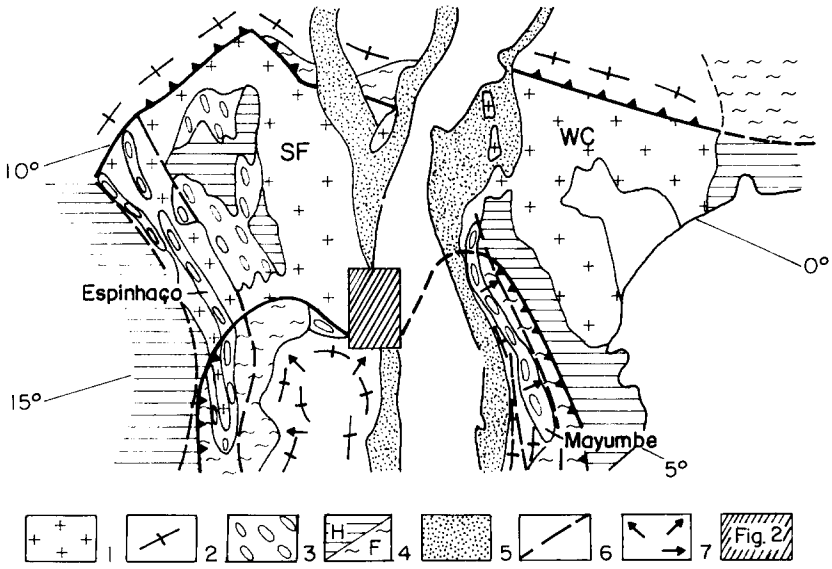


Fig. 1. Late Proterozoic geotectonic outline of correlative fragments from Brazil and Africa (adapted from [6,7]). *1* = Cratons: *SF* = São Francisco, *WC* = West Congo; *2* = Late Proterozoic marginal belt; *3* = Middle to Late Proterozoic intracratonic belt and associated cover; *4* = Late Proterozoic Platform Cover: Horizontal (*H*), Folded (*F*); *5* = Mesozoic/Cenozoic; *6* = Cratonic limits; *7* = Tectonic vergence shown by the Araçuaí–West Congo belts.

The area comprises medium to high-grade terranes (mainly charnockitic granulites, granites and migmatites) which extend along the Atlantic coast of Bahia and show submeridian (N–S) structural/lithologic trends (Fig. 2). They include an Early Proterozoic belt (2.4–2.0 Ga) with some older granulite remnants, and which borders the Archean Jequié complex [9,11–14].

The Sr and Ar systems of these metamorphic rocks from the coastal region (and from the Jequié Complex outside of the area) were homogenized and reset during the Trans-Amazonian cycle (2.2–1.9 Ga) as a result of the intense crustal reworking with granitization and migmatization events and the magmatic intrusion of juvenile rocks (e.g. [15,16]).

Several extensive, regional lineaments are identified in the investigated area (Fig. 2). They are oriented in NE–SW and NW–SE directions and cut the oldest submeridian structural trend of the Early Proterozoic mobile belts. Mafic dykes of the region may be associated with such regional structures but especially along the coast (between Ilhéus and Olivença—Fig. 2) they strike approximately perpendicular to the coastline (N 70° E to N 100° E) and are generally subvertical. Tectonically, these mafic dykes have been associated with extensional processes perhaps related to the evolution of the Espinhaço system (e.g. [16]). They are basic in composition, have subophitic textures and locally show minor secondary alteration (e.g., sericitized plagioclase). The dykes consist primarily of plagioclase (generally fresh, optically zoned) and pyroxene (Plate 1,2). Traces of brown biotite and dark green amphibole are common deuteric accessories in coarse-grained dyke interiors. The ubiquitous oxides are discussed in greater detail below.

3. Procedure

A total of 199 oriented block samples were collected from 66 dykes exposed along beach outcrops between Ilhéus and Olivença. Seventeen samples were also collected from the basement rocks near the contact of three dykes in the Olivença area for baked contact studies (sites 18b, 21b, 24b in Table 1). The samples were located less than 6 meters from the contact of each dyke. Ninety-two oriented block samples were also collected from 32 dykes exposed along the BR-101 highway between Camacã and Itabuna and along roads and quarries around the towns of Santa Cruz da Vitória, Itajú do Colônia and Palmira, which are referred to collectively hereafter as the Itajú do Colônia area (Fig. 2).

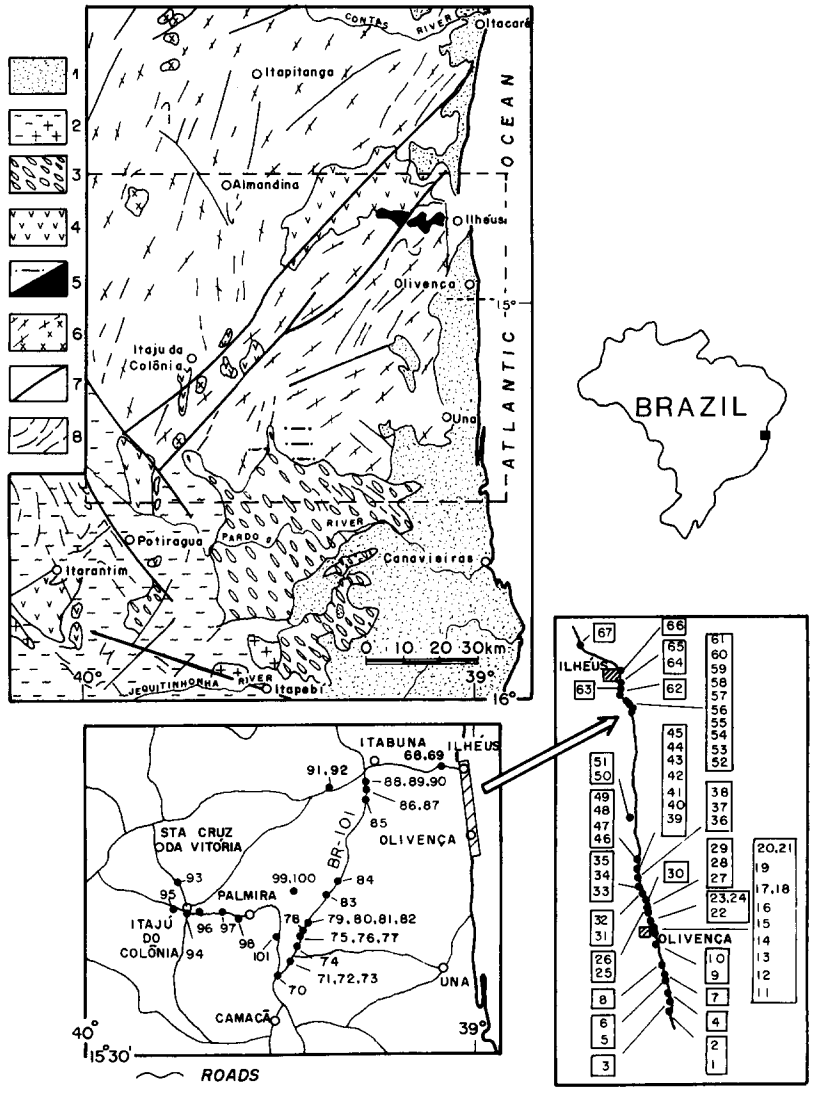


Fig. 2. Geotectonic/geologic map of the investigated area (adapted from [7,11]). *1* = Cenozoic; *2* = Late Proterozoic marginal belt (0.74–0.66 Ga): migmatites, gneisses, granites (+); *3* = Late Proterozoic Rio Pardo cover (> 0.55 Ga); *4* = Alkaline complexes (0.66–0.65 Ga); *5* = Diabases, gabbros; *6* = Early Proterozoic belt (2.4–2.0 Ga): medium-to high-grade terranes, granites (×); *7* = Main faults; *8* = structural trends. Paleomagnetic sampling sites are shown in the lower part of the figure. Each number represents a sample site.

Generally, three block samples spaced at least one half meter apart were collected from each dyke (considered as a sampling site). Whenever possible, both solar and magnetic compasses were used for orienting samples. For the majority of the samples the agreement of these measurements was good, the difference between them being generally less than 6°. In many cases, textural differences are evident between different blocks within a dyke due to differential cooling rates. Three to four specimens 2.5 cm in diameter and height were cut from cores extracted from each hand sample, and these were submitted to both alternating field (AF) and thermal treatments. A spinner magnetometer was used for the remanent magnetization measurements. Low demagnetization levels typically removed secondary components with low coercivities (< 150 mT) which were either randomly oriented or corresponded to the present geomagnetic field. Thereafter, stable directions

TABLE 1

Site mean paleomagnetic results

Site (No.)	Sample (No.)	Strike	Width (m)	N	Site mean direction				n	Virtual geomagn. pole	
					Dec. (°)	Inc. (°)	κ	α_{95}		Long. (°E)	Lat. (°N)
Olivença											
1	588–590	N80W	10	3	60.9	−44.5	59.8	32.8	2	72.8	32.3
2	591–594	N80W	20	4	59.9	−35.0	38.1	41.5	2	64.5	32.7
3	595–597	N80W	0.5	3	263.0	57.2	–	–	1	266.8	−14.5
4	609–611	N60E	1	3	140.1	−73.8	448.8	5.8	3	122.0	−8.6
5	612–614	N80E	1.5	3	114.2	−61.8	101.9	25.0	2	98.8	−6.5
6	615–617	N80E	20	3	129.1	−58.6	265.5	7.7	3	101.9	−17.9
7	618–619	N80W	1.5	2	61.8	−63.9	478.1	11.4	2	95.5	30.3
8	598	N65E	0.3	1	184.3	−67.2	–	–	1	144.0	−24.9
9	599–600	N80W	0.4	2	41.7	−81.9	88.6	26.8	2	129.3	26.3
10	601–603	N75W	5	3	87.3	−64.1	127.7	22.2	2	95.5	12.5
11	256–258	N70E	20	3	64.9	−64.1	541.1	5.3	3	105.3	28.0
12	262–264	N70W	0.5	2	334.0	62.3	–	–	1	300.1	26.7
13	259–261	N82W	1.2	3	312.9	56.0	64.3	15.4	3	281.5	21.9
14	252–255	N75E	1.0	3	330.6	52.6	17.3	30.5	3	291.1	34.2
15	246–248	N70E	1.5	3	283.8	53.8	31.8	22.2	3	267.5	2.5
16	249–251	N70E	0.4	3	304.5	58.7	45.8	37.7	2	279.7	14.9
17	239–241	E–W	1.0	3	296.8	32.1	–	–	1	256.1	19.7
18b	242–245	–	–	4	292.6	51.8	28.2	17.6	4	268.7	10.1
19	234–237	E–W	1.8	4	313.3	60.0	213.1	17.1	2	285.3	19.3
20	227–229	E–W	0.4	3	278.1	76.8	14.5	33.6	3	295.6	−10.1
21b	232–233	–	–	4	267.2	72.9	39.6	40.7	2	288.3	−14.1
22	273–275	N80E	0.8	3	299.3	65.1	79.4	13.9	3	284.2	7.6
23	265–267	N85E	15	3	314.4	63.2	86.0	13.4	3	288.8	17.4
24b	268–272	–	–	5	294.7	61.5	182.8	5.6	5	278.6	7.0
25	276–278	N85E	3	3	291.5	60.4	35.3	21.0	3	276.4	5.4
26	279–281	N70W	1.5	3	321.7	61.6	268.2	7.5	3	291.4	22.4
27	282–284	E–W	8	3	312.0	57.4	17.4	30.4	3	282.2	20.4
28	285–287	N60E	2	3	54.9	−67.4	145.9	10.2	3	102.0	33.6
29	288–290	N65E	2.5	3	72.8	−71.6	28.7	23.4	3	106.2	21.9
30	291–293	E–W	3.5	3	93.3	−74.2	269.5	7.5	3	110.8	11.3
31	294–296	N75E	1.0	3	72.5	−69.5	458.0	5.7	3	102.8	22.3
32	297–299	N80E	2.0	3	89.5	−71.9	171.7	9.4	3	106.8	12.7
33	301–302	N80W	1.0	3	249.6	54.4	275.5	15.1	2	262.9	−25.0
34	303–305	N55E	15	3	74.2	−76.6	64.3	15.4	3	114.8	20.2
35	306–308	N80E	1.0	3	56.9	−70.6	3873.4	1.9	3	106.7	30.9
36	451–452	–	10	2	69.1	−68.8	47.1	37.1	2	102.1	24.5
37	453–456	N85W	0.6	4	26.7	−77.3	103.3	9.0	4	116.6	22.2
38	457–459	N85E	0.4	3	81.3	−68.7	30.3	22.7	3	101.5	17.0
39	461–463	N65E	2.0	3	195.9	−80.7	63.6	15.6	3	145.8	−2.4
40	464–466	N80E	3.0	3	100.3	−72.2	500.0	5.5	3	108.6	7.1
41	467–469	N80E	6.0	3	94.9	−69.7	609.7	4.9	3	104.1	9.1
42	470–472	E–W	1.0	3	90.2	−73.0	254.2	7.7	3	108.6	12.6
43	489–491	N70W	2.0	3	121.8	−60.4	1527.9	3.0	3	100.0	−12.2
44	485–488	N70W	7.0	4	–	–	–	–	–	–	–
45	481–484	N85W	10	4	91.9	−65.6	127.7	8.1	4	98.0	9.7
46	492–494	N70W	–	3	42.0	−75.6	162.0	9.7	3	119.4	33.7
47	495–497	N50E	20	3	57.0	−50.2	414.7	6.0	3	78.7	35.6
48	498–499	N85E	1.5	2	–	–	–	–	–	–	–
49	500–502	N75E	6.0	3	134.5	−70.1	–	–	1	115.7	−10.9
50	478–480	N20W	20	5	32.6	−76.5	178.9	18.7	2	124.3	35.5
51	473–477	N20W	10	3	50.3	−67.2	913.2	4.0	3	103.0	36.3

TABLE 1 (continued)

Site (No.)	Sample (No.)	Strike	Width (m)	N	Site mean direction				n	Virtual geomagn. pole	
					Dec. (°)	Inc. (°)	κ	α_{95}		Long. (°E)	Lat. (°N)
Ilhéus											
52	421–424	N80W	10	4	65.6	−60.6	76.0	10.6	4	90.5	27.9
53	425–427	E–W	3	3	45.4	−65.2	812.7	4.3	3	101.6	40.3
54	428–430	N80E	8	3	42.3	−70.8	53.1	17.0	4	111.6	38.1
55	431–433	N80E	1	3	53.3	−64.8	1370.5	3.3	3	98.5	35.5
56	434–436	N80E	2	3	51.0	−65.6	83.7	13.5	3	100.3	36.6
57	436A–438	N80E	6	3	49.1	−65.4	211.4	8.4	3	100.7	37.9
58	439–441	N75E	0.5	3	63.5	−63.5	293.2	7.2	3	94.8	29.0
59	442–444	E–W	2	3	59.9	−60.9	99.4	12.4	3	91.4	32.0
60	445–448	N80E	1.5	4	54.4	−64.2	98.7	9.2	3	97.3	35.0
61	499–450	E–W	2	2	52.3	−63.4	1324.1	6.8	2	96.9	36.7
62	406–408	E–W	4	3	73.8	−70.2	774.8	4.4	3	103.9	21.3
63	418–420	N75W	1	3	60.4	−71.4	514.0	5.4	3	107.4	28.5
64	415–417	N80E	1.5	3	79.5	−73.2	225.2	8.2	3	108.7	18.0
65	412–414	N80E	1.5	3	73.3	−66.5	1080.4	3.7	3	98.2	21.9
66	409–411	N50E	4	3	58.2	−63.9	230.8	8.1	3	96.0	32.5
67	604–606	N85W	6	3	–	–	–	–	–	–	–
68	309–311	–	–	3	65.0	−68.0	1511.7	3.1	3	101.1	27.0
69	312–314	–	–	3	85.6	−71.7	90.7	13.0	3	106.3	14.7
Itajú do Colônia											
70	526–528	N32E	–	3	233.5	−21.7	104.8	12.1	3	207.0	−30.8
71	532–534	N40E	5	3	–	–	–	–	–	–	–
72	529–531	N40E	30	3	–	–	–	–	–	–	–
73	345–347	N60W	0.2	3	87.8	−68.8	374.8	6.3	3	101.5	13.3
74	535–538	N–S	10	4	87.4	−61.4	195.7	6.6	4	91.7	12.1
75	539–541	N40W	3	3	–	–	–	–	–	–	–
76	542–543	N40W	1	2	281.2	71.0	79.6	28.4	2	286.5	−6.3
77	544–546	N40W	15	3	–	–	–	–	–	–	–
78	547–550	N40E	–	4	109.4	−53.4	14.8	33.2	3	88.9	−6.9
79	551–553	E–W	20	3	116.7	−62.7	–	–	1	100.2	−7.4
80	554–556	E–W	15	3	86.4	−56.1	37.8	6.7	3	85.7	11.8
81	557–559	E–W	6	3	282.2	59.9	–	–	1	272.8	−0.9
82	560–561	E–W	1	2	110.0	−69.2	744.9	9.2	2	105.9	0.5
83	562–564	E–W	4	3	107.4	−74.9	55.9	16.6	3	113.6	5.3
84	565–568	N80E	30	4	74.6	−74.2	131.9	10.8	3	110.1	20.6
85	575–574	N60E	1	3	156.3	−75.3	–	–	1	129.8	−10.6
86	575–577	N30E	25	3	272.3	74.8	41.2	19.5	3	291.5	−11.9
87	578–579	N80W	0.5	2	205.4	68.7	–	–	1	297.7	−47.7
88	580–582	N80W	3	3	274.0	61.9	316.4	6.9	3	273.5	−7.2
89	583–585	N80W	3	3	–	–	–	–	–	–	–
90	586–587	N80W	2	2	247.4	60.5	–	–	1	270.0	−26.6
91	315	–	–	1	284.7	−81.5	–	–	1	157.6	18.3
92	316–318	N80E	1	3	285.5	−79.1	3045.4	2.2	3	162.1	19.3
93	319–321	N40E	0.4	3	87.9	−77.4	62.0	15.7	3	115.3	14.5
94	322–324	N40W	1.2	3	274.9	53.7	17.6	63.4	2	304.2	4.5
95	325–327	N50E	3	3	72.4	−68.2	150.9	10.0	3	100.0	22.7
96	328–330	N15W	1.2	3	300.3	77.2	262.5	15.4	2	299.3	−2.0
97	331–333	N60E	2	3	–	–	–	–	–	–	–
98	334–335	N65W	0.6	2	–	–	–	–	–	–	–
99	336–338	N30W	2	3		–	–	–	–	–	–
100	339–341	N80W	1.5	3	337.2	73.7	74.6	29.3	2	309.0	12.8
101	342–344	N20W	3	3	337.7	59.3	1669.4	6.1	2	300.7	30.9

Sample No.—Laboratory No.; Strike—attitudes are all subvertical; Width—dyke width in meters; N—number of collected samples; n—number of samples used for mean determinations; b—basement rocks.

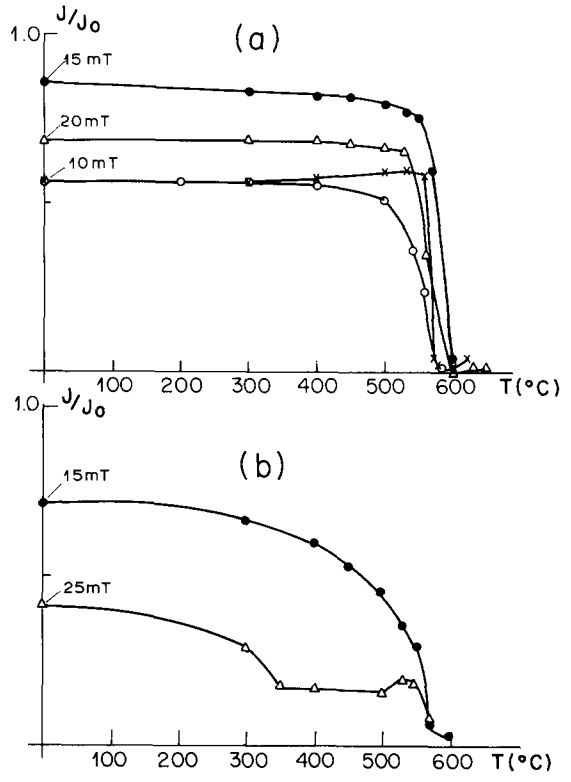


Fig. 3. Examples of normalized intensity curves: (a) samples BH-316-A (site 92) (full circles) and BH-535-B1 (site 74) (crosses) from Itajú do Colônia; BH-406-B1 from site 62 in Ilhéus (empty triangles), and BH-292-A1 from site 30 near Olivença (empty circles); (b) samples BH-319-C1 (site 93) (full circles) and BH-326-B1 (site 95)(empty triangles) from Itajú do Colônia.

(either normal or reversed) could be isolated for most dykes which presented unblocking temperatures ranging between 450 and 600 °C although a few showed distributed blocking temperatures (Fig. 3). The directions reported in table 1 represent the average of sample directions yielding the best cluster after combined AF and thermal demagnetization. After thermal treatment at 600 °C the sample magnetization intensities were too weak for consistent results to be obtained. Thermal magnetic curves (saturation magnetization measurements as a function of temperature) performed in air (Fig. 4) for several samples show similar behavior, with Curie temperatures between 570 and 600 °C, suggesting that magnetite is probably the main magnetic carrier in these samples.

4. Paleomagnetic results

Although the general behavior of all samples is similar, minor differences in the magnetic behavior between the different swarms of dykes are evident. These differences may reflect systematic variations in petrologically determined primary rock-magnetic features, and in the degree of weathering (extent of secondary overprints).

4.1. Ilhéus region

The samples from 17 dykes in the Ilhéus region yielded NRM intensities ranging from 1.0 A/m to 50 A/m and respond stably to AF treatment. Generally, after demagnetization up to 10–20 mT, a secondary component was eliminated leaving a stable easterly direction with relatively high upward inclination (Fig. 5a,b,c). The median destructive fields (MDF) of these samples are variable, with values smaller than 10 mT to values as high as 70–80 mT. Thermal treatment was also employed for these samples, after removing the low-coercivity secondary component with AF demagnetization (Fig. 5d,e). Generally, the samples

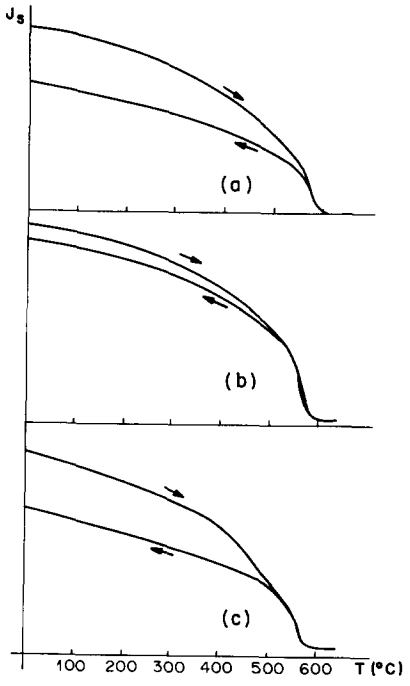


Fig. 4. Thermomagnetic curves: (a) sample BH-223-B from site 14 near Olivença; (b) sample BH-234-B from site 19 near Olivença; (c) sample BH-311-D from site 68 west of Ilhéus.

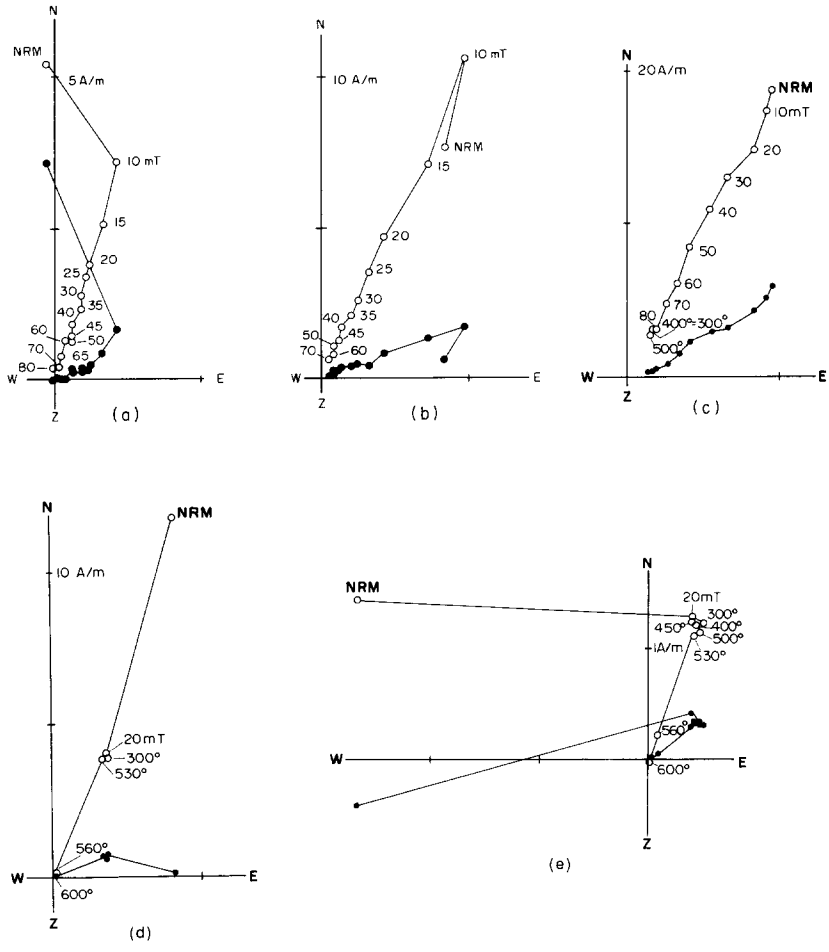


Fig. 5. Zijderveld diagrams for Ilhéus samples. Open (full) circles are vertical (horizontal) projections. (a) sample BH-413-A1 (site 65); (b) sample BH-440-A1 (site 58); (c) sample BH-448-A1 (site 60); (d) BH-440-B1 (site 58); (e) BH-431-B1 (site 55).

presented the same behavior as in the AF treatment, but for some dykes there was an improvement in the clustering of directions. One dyke, site 67, behaved unstably with respect to AF demagnetization and is not included in the statistical analyses.

4.2 Olivença region

Samples from the 48 dykes collected along the Olivença beach (Fig. 2) have NRM intensities ranging from 10 A/m to 160 A/m and both polarities were found after AF and thermal treatments. Two dykes, sites 44 and 48, did not yield stable remanent magnetizations, and the remaining samples presented a somewhat different response to AF treatment, compared with that from dykes of the Ilhéus region as described below.

The MDF of these samples rarely exceeds 10 mT and the intensity of remanence often decreases to less than 10% of the NRM for this demagnetizing field. Several samples (mainly those with downward inclinations) behaved unstably after demagnetization to 15–20 mT, and the thermal treatment did not alter significantly the NRM directions. However, when specimens were AF demagnetized to 10–15 mT and then thermally demagnetized, stable directions were isolated for several samples (Fig. 6a,b—downward inclinations). This behavior probably indicates that the AF treatment up to 10–15 mT eliminates substantially the secondary directions and decreases the intensity of remanence to values comparable to that of the characteristic magnetization remaining in the rocks. Subsequent thermal treatment then

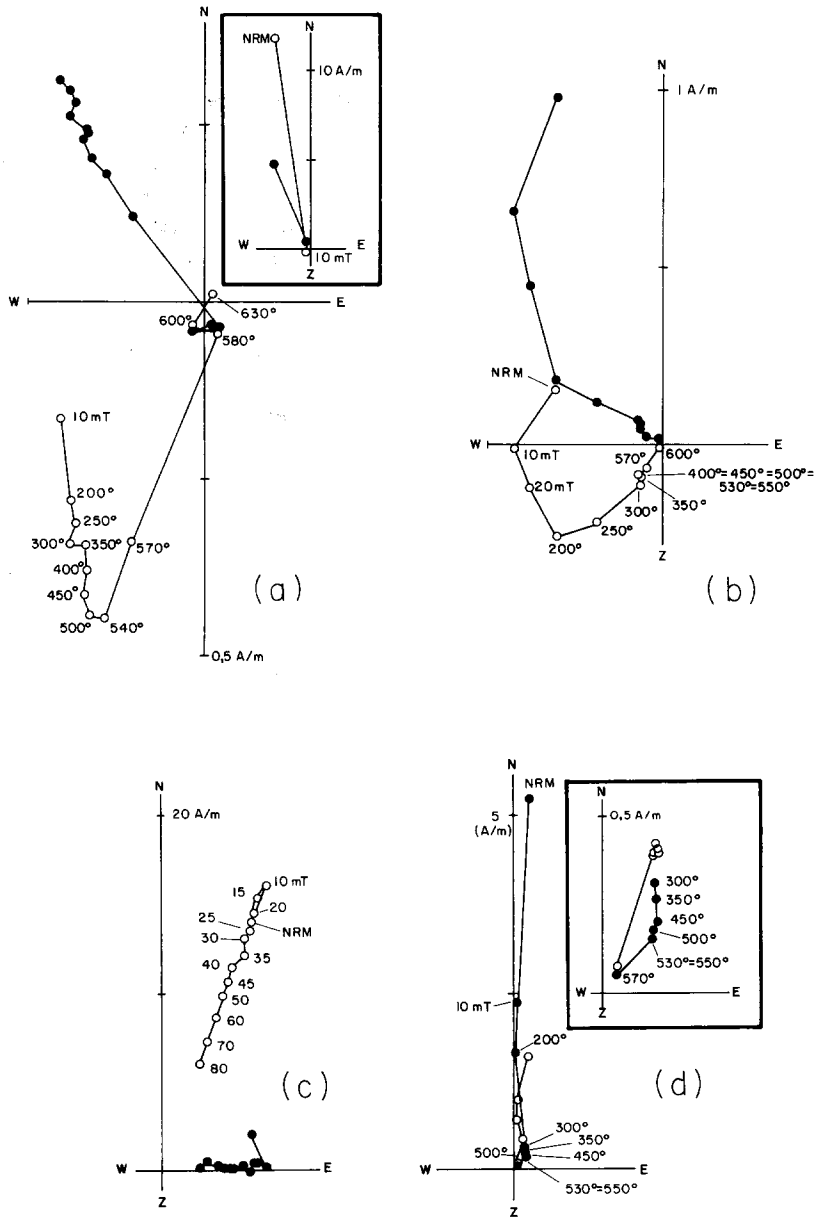


Fig. 6. Zijderveld diagrams for Olivença samples. Open (full) circles are vertical (horizontal) projections. (a) sample BH-236-B1 (site 19); (b) sample BH-249-C1 (site 16); (c) sample BH-467-A1 (site 41); (d) sample BH-287-B1 (site 28).

becomes more effective in separating the components (e.g., [17]). Two samples for which upward inclination directions were recovered are BH-467-A1 which are stable to fields up to 80 mT (Fig. 6c), and BH-287-B1 for which the mixed treatment was employed (Fig. 6d). Generally, the less stable secondary directions were random with the exception of three dykes in which two directions with nearly opposite polarity could be isolated (Fig. 6a). These samples presented steep negative inclination NRM's similar to the normal polarity directions obtained for the Ilheus and Olivença dykes, but after AF demagnetization followed by thermal treatment a westerly direction with high downward inclination could be isolated. The upward component in these sample has distributed

unblocking temperatures, ranging from < 200 to ~ 500 °C.

All samples of basement rocks and of an older dyke, site 15, collected within one meter of younger discordant dykes yield the same directions as the basic dykes after AF and thermal treatment. A stable remanent direction representative of the unreheated metamorphic basement or older dyke could not be determined from the present sample collection.

4.3. Itajú do Colônia region

The NRM intensities of these samples range from 5 to 10 A/m and responded less stably to AF and thermal treatments than do those samples from Ilhéus and Olivença. Samples from 8 sites behaved unstably and did not yield consistent results. Of the remaining 24 dykes, 13 yielded an easterly direction with high negative inclination (Fig. 7a) and 10 yielded a westerly direction with steep downward inclination (Fig. 7b), although for some of them only one or two samples (from three possibilities) gave acceptable results (Table 1). Finally, one site (site 70 of Table 1) also presented good stability to AF and thermal treatment, but the isolated magnetization had a southwesterly direction with moderate upward inclination (Fig. 7c) and was not included in the calculation of the average directions.

After elimination of secondary directions in fields of 10–15 mT, several samples became unstable to greater fields, although good stability to fields as high as 80 mT was also found. The MDF of these samples averaged approximately 5–10 mT, ranging to values as high as 30–40 mT. As for the Olivença dykes, the mixed treatment was also the most effective in isolating the components present in several rocks.

5. Petrography of Fe-Ti oxides

Polished thin sections of several dykes from each area (except Itajú do Colônia) were examined petrographically and with a scanning electron microscope (SEM) to establish the magnetic mineralogy. A magnetic colloid [19] was used in conjunction with optical properties, backscattered electron imaging, and semi-quantitative energy dispersive X-ray analysis to identify different phases.

All samples examined contain coarse grains (up to ca. 600 μm) of intergrown magnetite and ilmenite, the latter commonly occurring in both coarse equant subgrain domains and in relatively thin lamellae (Plate 1b, upper right) parallel to

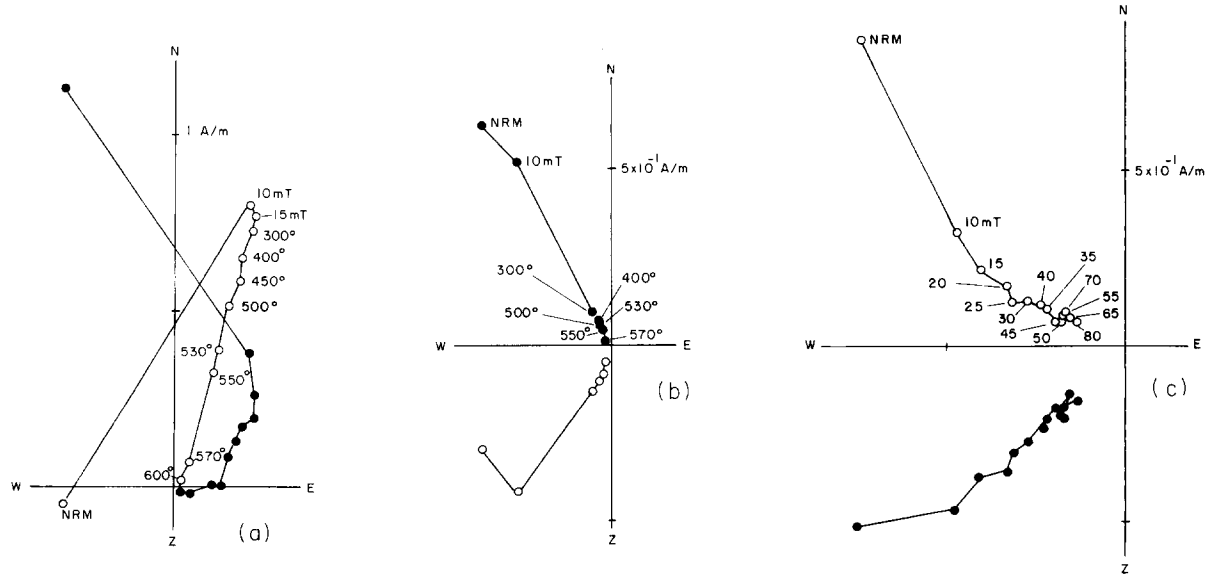


Fig. 7. Zijderveld diagrams for Itajú da Colônia samples. Open (full) circles are vertical (horizontal) projections. (a) sample BH-319-C1 (site 93); (b) sample BH-343-C1 (site 101); (c) sample BH-526-A1 (site 70).

{111} in the host magnetite. These features are inferred to result from high-temperature oxidation/exsolution upon initial cooling of the dykes. Within the coarse ilmenite subgrain domains, intergrowths of an oxide with a rodlike habit frequently occur. The rods are less than 1 $\mu m \times 10$ μm, and are occasionally large enough to be resolvable with the optical microscope, in which case their reflectivity and attraction of the colloid serve to identify them as magnetite. More often, these rods were recognizable only with the SEM (Plate 1,c), whereupon at least two directions of preferred orientations of the rods is visible. The habit and preferred orientation of the magnetite rods is similar to those reported [20–22] in silicate minerals, but has not, to our knowledge, been noted in ilmenite before. The observation of magnetite within the size range for which single domain behavior is predicted [23] explains the stability of the remanence to AF demagnetization.

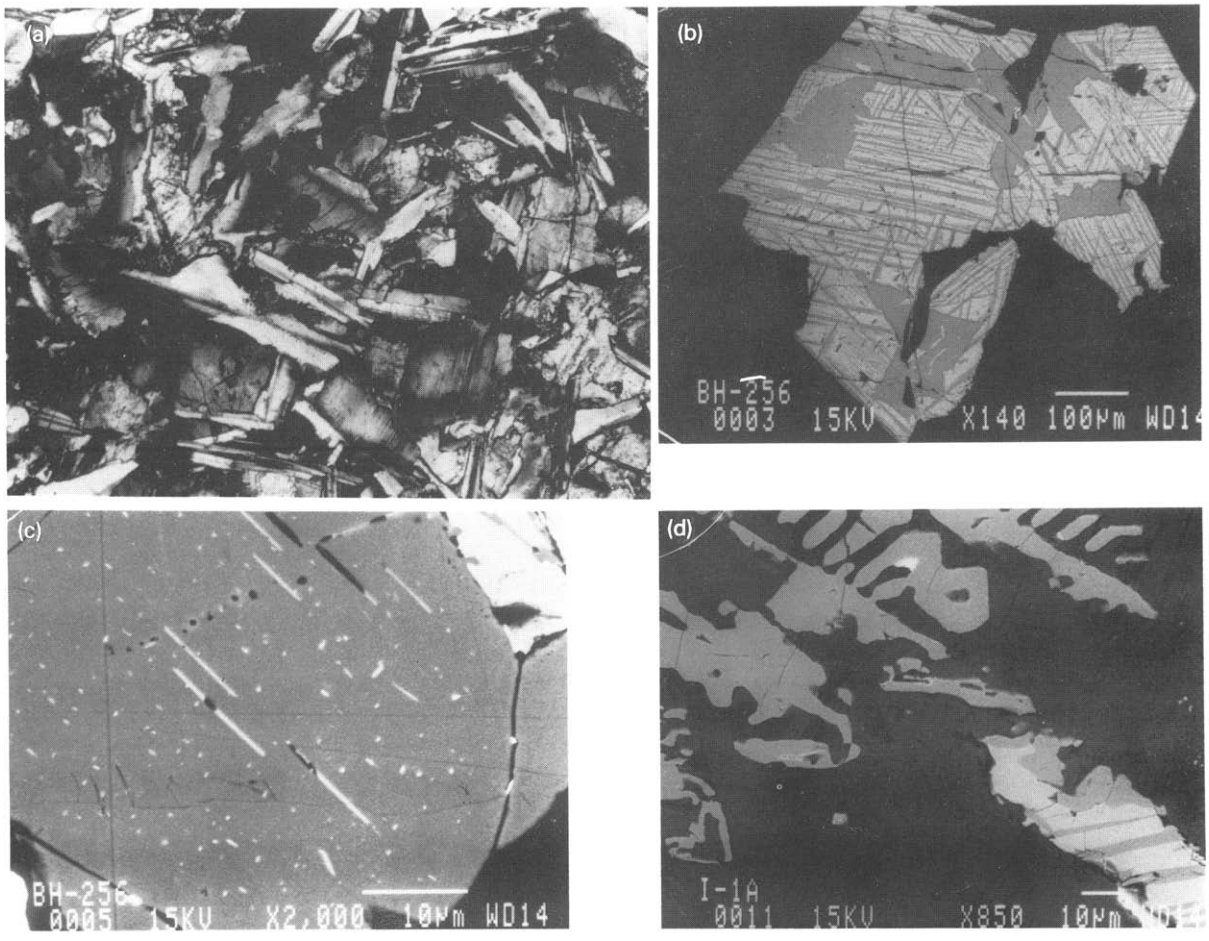


Plate 1. (a) Transmitted light (crossed polars) photomicrograph of sample BH-252 (site 14, group O_R). Tabular crystals (laths in cross section) of fresh, optically zoned plagioclase and interstitial clinopyroxene define a sub-ophitic texture. Horizontal dimension of photo is 2.5 mm. (b) Backscattered electron SEM image of composite oxide grain from sample BH-256 (site 11, group O_N). Darker patches and fine lamellae are ilmenite, brighter regions are magnetite. Ilmenite lamellae are oriented along {111} planes in the host magnetite, and are interpreted to result from oxidation/exsolution upon initial cooling of the dyke. Scale bar is 100 μm. (c) Backscattered electron SEM image of one of the ilmenite patches (grey) within a larger exsolved oxide grain in sample BH-256 (site 11, group O_N), showing a finer scale exsolution of magnetite rods (white). Note two preferred orientations of the magnetite rods, one nearly parallel to and one nearly perpendicular to the image plane. Scale bar is 10 μm. (d) Backscattered electron SEM image of skeletal ilmenite crystals (upper left) and a composite ilmenite/magnetite grain (striped grain, lower right) in a sample from site 62 of Table 1 (group I_N). Scale bar is 10 μm.

In addition to the magnetite-ilmenite intergrowths, some dykes contain discrete ilmenite occurring as skeletal, dendritic or trellis-like crystals (Plate 1,d). Relatively large networks of these crystals have indistinguishable optical orientation, therefore three-dimensional crystallographic continuity appears likely. These ilmenites are homogeneous at the scale of SEM resolution, and appear compositionally similar to the ilmenite intergrown with magnetite.

Pyrite is present in most samples, generally as discrete grains but rarely (in one sample) also as microscopic veins. Less common, but often associated with the pyrite, is pyrrhotite. Patchy adherence of the magnetic colloid to pyrrhotite grains may reflect magnetocrystalline domain structure of this phase, or may indicate a range of magnetic properties resulting from compositional heterogeneity. Petrographic features of both pyrite and pyrrhotite suggest a deuteric or supersolidus origin.

The optical observations described herein aid in the interpretation of the magnetic properties evidenced by the AF and thermal treatments. The unblocking temperatures associated with the characteristic magnetization of these rocks are consistent with magnetite as the main carrier of remanence (Fig. 3a). Also, a perceptible drop in the intensity of magnetization around 350 °C observed for some samples (e.g., Fig. 3b) accords with the observed presence of pyrrhotite. The coarse subgrains of magnetite (probably with multidomain magnetocrystalline structure) observed in the thin sections, could be responsible for the unstable behavior presented by many samples in the AF treatment while the small rodlike subgrains of magnetite (probably having single-domain structure) observed in the ilmenite subgrains, could be associated with the stability of the characteristic component.

6. Discussion

The average direction for each polarity population of the Olivença dyke swarm are not symmetrically opposed (Fig. 8a), whereas those of the dykes from Itajú do Colônia are (Fig. 8c). A spatial polarity pattern is also apparent when the inclination and declination of the stable remanent directions yielded by the 66 dykes that occur along the coast from Olivença to Ilhéus are plotted

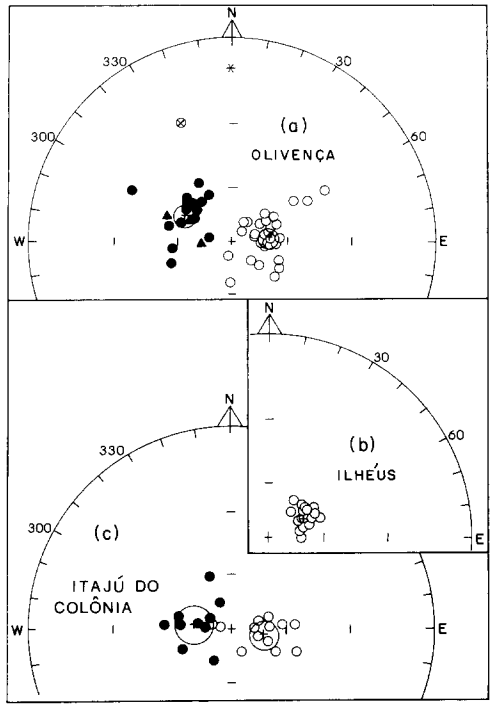


Fig. 8. Site mean directions. (a) Olivença dykes; (b) Ilhéus dykes; (c) Itajú da Colônia dykes. Open (full) circles represent upward (downward) inclinations. Triangles denote basement rocks close to the contact and metamorphic dyke mean directions. The symbols + and * are the present geomagnetic field and the dipolar field, respectively.

against the present-day site latitude from south to north (Fig. 9). Disregarding the two single dykes with positive inclinations, three broad polarity intervals are discernable: two with negative inclinations and one with positive inclination.

Before using these paleomagnetic data to draw tectonic conclusions [5], several questions must be addressed such as whether or not the magnetizations are primary TRM's, whether or not the magnetizations have averaged geomagnetic secular variation, whether or not the geographically distinct swarms yield statistically different mean directions, and whether or not the magnetizations record a non-dipolar field.

6.1. Evidence for primary TRMs

The baked contact tests performed on some dykes are incomplete as no independent direction (i.e., older direction) could be found for the country rocks. However, paleomagnetic results from the Jequié Complex, approximately 100 km to

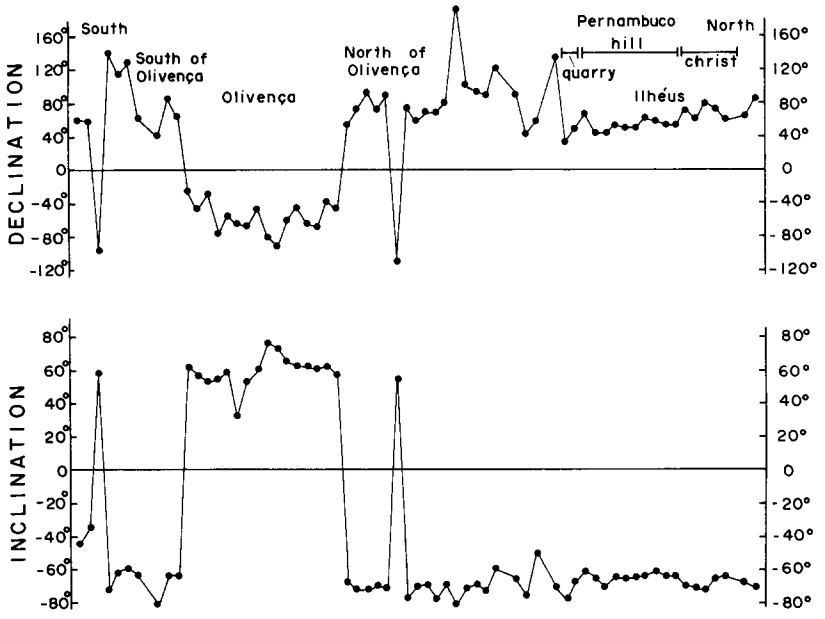


Fig. 9. Variation of declination and inclination of the dykes sampled along the coast between Olivença (South) and Ilhéus (North). The horizontal axis corresponds to sampling ordered from South to North.

NW, possess stable, northeasterly, positive inclination directions associated with ~ 2.0 Ga $^{40}Ar/^{39}Ar$ hornblende and biotite dates [18]. Since samples of the basement rock near the contact with the Olivença and Ilheus dykes also yield ~ 2.0 Ga $^{40}Ar/^{39}Ar$ hornblende dates [5], we suspect that the initial magnetization of the metamorphic basement was very similar to that of the Jequie Complex and quite distinct from that of the dykes. The similarity of the directions from the dykes and from the basement close to the contact suggests either that the country rocks were remagnetized during the dykes intrusion or that the studied area was submitted to a later remagnetization episode.

Because the dyke samples possess characteristic components with high unblocking temperatures and because petrographic and SEM observations indicate that some of this remanent component resides in single-domain size magnetite formed as a result of high-temperature oxidation/exsolution upon initial cooling of the dykes, then thermal remagnetization of the dykes' initial TRM would require a thermal episode of high peak temperature. Even if we accept 500 °C as a minimum unblocking temperature for the characteristic component, a peak temperature of at least 400 °C would have to be applied for 10 Ma to completely reset the magnetization [24]. Such temperatures for such durations would result in the formation of greenschist facies mineral assemblages, yet petrographic and SEM observations failed to detect any such metamorphic assemblages or even low-temperature alteration assemblages, which might be associated with a thermal or chemical magnetic overprint in either the dykes or the country rocks.

$^{40}Ar/^{39}Ar$ studies have shown that the only thermal disturbance in the region subsequent to the intrusion of the dyke swarms is that due to the 600–650 Ma Brasiliano episode, the peak temperature of which never exceeded 150 °C [5]. Clearly, the dykes' characteristic magnetizations are impervious to thermal overprinting by such a low-temperature event. The fact that none of the mean poles coincide with the Brasiliano age paleomagnetic pole from the Piquete region of the southern margin of the São Francisco Craton [25] further supports this conclusion. The spatial polarity pattern found for the Olivença–Ilhéus beach samples (Fig. 9) would also require an areally selective bipolar remagnetization process which seems highly unlikely.

The results from the baked contact samples, therefore are most consistent with the interpreta-

tion that the characteristic magnetizations of the dykes are thermoremanent in nature, acquired upon injection and crystallization of the dykes.

6.2. Secular variation

Rapid cooling of the dykes after injection [5] suggests that the magnetizations of individual dykes represent essentially a spot reading of the geomagnetic field, as cooling through magnetite blocking temperatures (580–500 °C) occurs over a time interval less than several years long. Thus a mean direction from a group of dykes may not completely average out secular variation. To test this the angular dispersion as calculated by two-tier analysis [26] of each group of VGPs presented in Table 1 is compared with the predicted dispersion due to secular variation [27] at the paleolatitude corresponding to the respective mean inclinations (Fig. 10). Only the dispersion for the mean pole *IC* of Table 2 is plotted as the two polarities comprising this group yield statistically indistinguishable mean directions and dispersion parameters (see below). All the calculated dispersions but one (Ilhéus-I_N) compares favorably with the values for the present-day field (Fig. 10) showing that the intrusion of dykes in both polarity groups probably occurred during a sufficiently long time interval to average secular variation, and that the mean direction for each of these group of directions reliably estimates the mean geomagnetic field orientation.

The Ilhéus pole (I_N—Table 2) may not completely average secular variation as its dispersion is much lower than those of the other dykes. On the other hand, thermal modelling of $^{40}Ar/^{39}Ar$ dates from the baked contacts of an Ilhéus dyke [5] suggest that the country rock temperatures may have been greater in the Ilhéus area than at Olivença at the time of dyke injection, which would favor protracted post-injection cooling rates for these dykes. In this case some averaging of secular variation would occur at the individual site (dyke) level and the resulting group mean would show less dispersion than predicted (Table 3). In any case, it will be shown (see below) that this pole is statistically different from the others at the 95% probability level and also, that if it is a

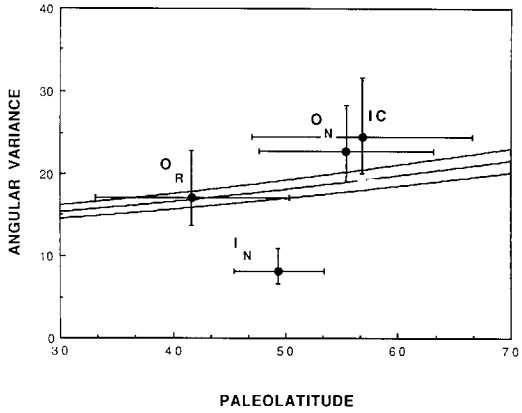


Fig. 10. Comparison of the dispersion of each group of VGPs presented in Table 1 with the predicted dispersion due to secular variation. Vertical axis plots angular dispersion, S. The middle solid curve represents the least squares best fit dipole/quadrupole model of [27] to IGRF65, where $S^2 = A^2 f^2 + B^2$. f is the latitude, $A = 0.24 \pm 0.04°$ and $B = 13.5 \pm 0.6°$ (2 s.d.). The upper and lower curves represent the bounding 95% error region. O_N, O_R, IC, and I_N represent the angular variances of Olivença normal polarity, Olivença reverse polarity, Itajú do Colônia combined polarities, and Ilheus normal polarity. The horizontal error bars are their α_{95}'s and the vertical error bars encompass the upper and lower 95% confidence values [28].

TABLE 2

Group mean paleomagnetic results

Locality	Sites (n)	Mean directions					Paleomagnetic pole				Magnetic age (Ma)
		D (°)	I (°)	κ	α_{95} (°)	R	Lat. (°)	Long (°)	K	A_{95} (°)	
Ilhéus-I_N	17	60.0	−66.8	189.7	2.6	16.92	30.3	100.4	78.9	4.0	1012 ± 24
Olivença-O_R	18	298.8	60.7	30.4	6.4	17.44	9.5	280.2	17.0	8.6	1078 ± 18
Olivença-O_N	31	82.4	−71.0	26.7	5.1	29.88	16.1	107.0	12.1	7.8	–
Itajú do Colônia-IC_R	10	280.3	70.1	24.9	9.9	9.64	−5.4	290.3	10.5	15.6	–
Itajú do Colônia-IC_N	13	97.7	−73.4	27.4	8.1	12.56	9.5	111.6	10.1	13.8	–
Itajú do Colônia-IC	23	99.0	−71.9	27.2	5.9	22.19	7.7	111.0	10.6	9.8	–

TABLE 3

Comparisons between experimental and predicted dispersions due to secular variation

Pole	Paleolatitude	Angular dispersion	95% limits
I_N	49.4±4.0	8.6	10.9– 6.5
O_R	41.7±8.6	18.4	22.8–13.7
O_N	55.4±7.8	23.0	28.0–19.2
IC	56.8±9.8	24.4	31.3–20.1

reliable pole it defines defines an APW path for the São Francisco Craton for the time interval between 1.01 and 1.08 Ga.

6.3. Distinctness of group mean directions

Although the geographical grouping of the dykes used herein yields groups of directions which probably average secular variation, their mean directions are similar (Table 2) and a statistical test [29] for discrimination of mean directions from Fisher [30] distributions must be applied to ascertain at what level of probability these mean directions may be considered distinct.

The hypothesis that two populations of samples N_1 and N_2, share a common true mean direction may be rejected at the level of confidence P, if:

$$\left[R_1 + R_2 - R^2/(R_1 + R_2)\right]/2(N - R_1 - R_2) > (1/P)^{1/(N-2)} - 1 \quad (1)$$

where R is the length of the vector sum of the resultants R_1 and R_2 of the separate populations and $N = N_1 + N_2$.

However, this statistical test is only valid for populations which have similar precisions ($\kappa_1 = \kappa_2$). If the two populations do not have a common precision parameter, the hypothesis of a common mean direction is rejected if:

$$r\left[(R_1 + R_2)^2 - R^2\right] / \{2[(N_1 - R_1) + r(N_2 - R_2)][R_1 + rR_2]\} > (1/P)^{1/(N-2)} - 1 \quad (2)$$

where $r = \kappa_2/\kappa_1$ which is estimated from the ratio κ_2/κ_1, with $\kappa_i = (N_i - 1)/(N_i - R_i)$. Since in this case the test involves an approximation, it should not be used to derive conclusions when both sides of expression (2) yield similar results.

Table 2 shows that the values of κ associated with the Olivença and Itajú do Colônia mean directions are similar and so expression (1) may be used for comparisons between them. However, the Ilhéus mean direction (I_N), has a significantly larger κ and expression (2) must be used whenever this mean direction is compared with others.

The reversed and normal mean directions from the Itajú do Colônia are statistically indistinguishable at the 95% probability level (Table 4). Thus a mean direction including both polarities for the Itajú do Colônia (IC—Table 2) is considered to be representative of the magmatic event in this region and will be used for a comparison with the other mean directions.

The differences between the mean directions are primarily in declination (Table 2), with the following sequence: $D = 60.0°$ for Ilhéus-I_N, $D = 82.4°$ for Olivença-O_N, $D = 99.0°$ for Itajú do Colônia-IC and $D = 298.8°$ (or 118.8° for comparison) for Olivença-O_R. Thus, comparisons between adjacent mean directions within the sequence (I_N and O_N, O_N and IC and IC and O_R) are sufficient to establish whether a particular

TABLE 4

Comparisons of mean directions

Pole 1	Pole 2	R	N	$r = \kappa_2/\kappa_1$	U	V	$(1/P)^{1/(N-2)} - 1$ ($P = 0.05$)	$(1-P)\times 100$ (%)
IC_R	IC_N	22.191	23	–	0.01250	–	0.15333	23
I_N	O_N	46.658	48	0.1407	–	0.1724	0.06729	100
O_N	IC	52.012	54	–	0.02859	–	0.05930	77
IC	O_R	39.357	41	–	0.19996	–	0.07984	100

The last column presents the probability of distinctness $[(1/P)^{1/(N-2)} - 1 = U$ (or $V)]$.

$U = [R_1 + R_2 - R^2/(R_1 + R_2)]/2(N - R_1 - R_2)$

$V = r\{(R_1 + R_2)^2 - R^2)/(2[(N_1 - R_1) + r(N_2 - R_2)](R_1 + rR_2)\}$

mean direction is statistically different from the others. Table 4 indicates that the Ilhéus-I_N and Olivença-O_R mean directions are statistically different from each other and from the other mean directions at the 95% probability level. The Olivença-O_N and the Itajú do Colônia-*IC* mean directions are statistically indistinguishable at the 95% probability level, but they are different at the 75% probability level. As these two groups of directions belong to different areas, they will be considered as being independent mean directions in the next topics.

6.4. Non-dipole field?

The presence of both polarities indicates that Bahia dyke intrusion occurred over a time scale long enough to span at least one geomagnetic field reversal. Departure from exact antipolarity of the two populations of directions found for Olivença basic dykes however may reflect: (1) a time lapse encompassing continental drift between the injection of the two groups; (2) secondary magnetizations were not completely eliminated for the samples which present positive inclinations (the remanence of these samples is unstable to AF treatment and the thermal treatment may not have been effective to completely eliminate secondary magnetizations); or (3) a long-standing non-dipole field component affecting magnetizations of possibly very similar ages.

Asymmetrical polarities have been reported for 1.1 Ga Precambrian rocks from the Lake Superior region [31]. However, this polarity asymmetry is mainly due to inclination anomaly while our polarity asymmetry is mainly due to a declination anomaly (Fig. 8b). A non-dipole field hypothesis is also unlikely as the bipolar magnetizations of the Itajú do Colônia basic dykes (Fig. 8c) are perfectly antipolar within error. Magnetizations from slowly cooled granulites of the Namaqua zone of the Kalahari Craton, dated at ca. 1.0 Ga [5] are also perfectly antipolar within error [32]. Results of $^{40}Ar/^{39}Ar$ step-heating analysis of minerals (hornblendes, biotites, plagioclases) from the granulitic host rocks of two dykes with opposite polarities sampled at Ilhéus (upward inclinations) and Olivença (downward inclinations), respectively, (Table 2, [5]) support the first explanation, as the 65 ± 45 Ma age difference between the Olivença dykes (O_R) and the Ilhéus dykes (I_N) is ample time for significant plate motion to have occurred. Indeed, the paleomagnetic poles from

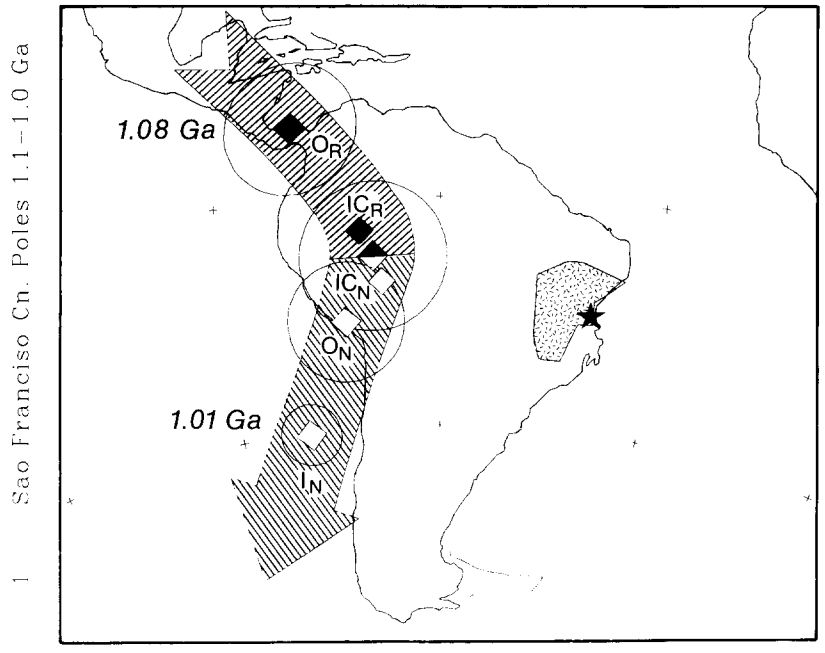


Fig. 11. APW path for the São Francisco Craton for the time interval 1.08–1.01 Ga defined by the poles of Table 2. Full (open) symbols represent dikes with downward (upward) inclinations. The star shows the sampling locality.

Table 2 define an APW path for the São Francisco Craton for this time interval (1012 ± 24 Ma–1078 ± 18 Ma, Fig. 11). Although only two of these poles (O_R and I_N) are dated directly, interpolation on the APW path suggests that O_R is older than O_N. With respect to this inference, three dykes from Olivença that yielded westerly, positive inclination, high-temperature (> 500 °C) component (group O_R) also possessed low-temperature (< 500 °C in Fig. 6a) directions of opposite polarity (group O_N). These group O_R dykes may have been partially remagnetized by the thermal anomaly associated with younger group O_N.

7. Conclusions

Statistical analysis of paleomagnetic data presented herein is consistent with episodic, local injection of the Bahia dykes in four pulses. Each of these pulses apparently occurred over hundreds or thousands of years, and much longer time intervals (10–20 Ma) elapsed between injection of the different pulses. Mean directions from dykes injected in each pulse define a time sequence whose end points are 65 ± 45 Ma apart in age.

The paleomagnetic poles appear to define an apparent polar wander path for the São Francisco Craton (Fig. 11) between the time interval 1012 ± 24 Ma (Ilhéus basic dykes -I_N) and 1078 ± 18 Ma (Olivença basic dykes -O_R). During this time interval at least two polarity intervals (reversed and normal) are in evidence. This APW path records a minimum drift rate of 2.8 cm/yr of the São Francisco Craton which is smaller than the inferred minimum plate velocities during Proterozoic time [33], but consistent with present-day plate velocities.

Finally, the paleomagnetic poles implied by steep paleomagnetic inclinations are consistent with evidence of glaciation in the São Francisco Craton at around 1.0 Ga [34,35]. Implications of the APW path defined herein to tectonic relationships of the São Francisco, Congo and Kalahari Cratons, and the origin of intervening Brasiliano/Pan-African mobile belts, are discussed by [5].

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